

OVAR-BPnet: A General Pulse Wave Deep Learning Approach for Cuffless Blood Pressure Measurement

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Abstract—Pulse wave analysis, a non-invasive and cuffless approach, holds promise for blood pressure (BP) measurement in precision medicine. In recent years, pulse wave learning for BP estimation has undergone extensive scrutiny. However, prevailing methods still encounter challenges in grasping comprehensive features from pulse waves and generalizing these insights for precise BP estimation. In this study, we propose a general pulse wave deep learning (PWDL) approach for BP estimation, introducing the OVAR-BPnet model to powerfully capture intricate pulse wave features and showcasing its effectiveness on multiple types of pulse waves. The approach involves constructing population pulse waves and employing a model comprising an omni-scale convolution subnet, a Vision Transformer subnet, and a multilayer perceptron subnet. This design enables the learning of both single-period and multi-period waveform features from multiple subjects. Additionally, the approach employs a data augmentation strategy to enhance the morphological features of pulse waves and devise a label sequence regularization strategy to strengthen the intrinsic relationship of the subnets' output. Notably, this is the first study to validate the performance of the deep learning approach of BP estimation on three types of pulse waves: photoplethysmography, forehead imaging photoplethysmography, and radial artery pulse pressure

waveform. Experiments show that the OVAR-BPnet model has achieved advanced levels in both evaluation indicators and international evaluation criteria, demonstrating its excellent competitiveness and generalizability. The PWDL approach has the potential for widespread application in convenient and continuous BP monitoring systems.

Index Terms—Blood pressure estimation, pulse wave analysis (PWA), pulse wave deep learning (PWDL), photoplethysmography (PPG), imaging photoplethysmography (iPPG), radial artery pulse pressure waveform (PPWr).

I. INTRODUCTION

BLOOD pressure (BP) serves as a crucial vital sign, offering insights into the overall health of the human body. Presently, cardiovascular disease remains the foremost global cause of mortality, with hypertension established as the primary contributor to cardiovascular issues [1]. The World Health Organization (WHO) estimates that approximately 1.28 billion adults aged 30–70 worldwide grapple with hypertension. Regrettably, 46% of these individuals are unaware of their condition. Meanwhile, only 42% have been both diagnosed and treated. In everyday life, the electronic sphygmomanometer typically takes precedence as the primary tool for individuals to measure their BP. Although this device uses a non-invasive cuff measurement method, it has certain limitations. One is that the measurement frequency is low, and the other is that using the cuff for too long may cause blood vessels to deform, thus affecting the accuracy of BP measurement [2]. Given that blood pressure is a dynamic physiological parameter, the traditional cuff BP measurement method falls short of fully capturing the subject's physiological state [3]. Consequently, the research of cuffless BP measurement technology emerges as of paramount significance for the routine health assessment of individuals.

In recent years, pulse wave-based cuffless BP estimation technology has become a research hotspot due to its convenient measurement and continuous monitoring capability [4]. The types of pulse waves can be divided into photoplethysmography (PPG), imaging photoplethysmography (iPPG), and radial artery pulse pressure waveform (PPWr) [5], [6], [7]. In addition, pulse wave detection technology uses sensitive photoelectric sensors or pressure sensors to record changes in arterial blood volume or pressure, thereby obtaining rich information about the

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cardiovascular system on the outer surface of the skin [8], [9]. Previous studies have shown that the morphology of pulse wave signals is strongly correlated with BP and contains potential BP information [10].

Currently, many researchers use the pulse wave analysis (PWA) method to correlate the waveform characteristics of pulse wave signals in one or more human body parts with BP [11]. PWA can meet our accuracy and continuity requirements for daily BP measurement [12]. Blood pressure mapping tools commonly used by researchers include machine learning algorithms and deep learning algorithms. In 2020, Hasanzadeh et al. estimated BP using PPG signals based on the adaptive boosting machine learning model. The mean absolute error (MAE) $\pm$ standard deviation (SD) of the estimated values of systolic blood pressure (SBP) and diastolic blood pressure (DBP) was 8.22 ± 10.38 mmHg and 4.17 ± 4.22 mmHg, respectively [13]. However, a typical problem is that the BP model established by machine learning algorithms relies heavily on the manual extraction of features [14]. Fortunately, deep learning algorithms can automatically extract waveform features from the original waveform and map them to the BP value. Therefore, researchers are paying more attention to the BP models based on deep learning [15], [16], [17].

In 2020, researchers, including Schlesinger, utilized the MIMIC-II database to establish a PPG dataset. They fed the signal into a convolutional neural network (CNN) architecture inspired by AlexNet. The estimated MAE $\pm$ SD for SBP and DBP were 7.34 ± 8.65 mmHg and 3.91 ± 4.48 mmHg, respectively [18]. In 2022, Wang et al. transformed the PPG signal, also sourced from the MIMIC-II database, into a visibility graph (VG). Subsequently, they employed a CNN model to extract feature vectors from the VG. The developed method achieved performance with an MAE of 6.17 mmHg and 3.66 mmHg for SBP and DBP. However, BP models based on a single CNN architecture predominantly focus on the spatial information of the pulse wave, neglecting the temporal information inherent in the time series signal [20], [21].

Consequently, recent research has delved into hybrid deep BP models, amalgamating CNN with other network structures or mechanisms to enhance estimation accuracy [22], [23]. In 2022, Paviglianiti et al. extracted PPG, electrocardiogram (ECG), and arterial blood pressure (ABP) from 40 patients in the MIMIC database [24]. The estimation model configuration involved a residual network (ResNet) and three Long Short-Term Memory (LSTM) networks. The MAE results for SBP and DBP using only PPG were 7.12 mmHg and 3.53 mmHg. In 2023, Li et al. introduced Hybrid D1DCnet, a BP model combining CNN, LSTM, and a self-attention mechanism. The MAE $\pm$ SD obtained on the iPPG dataset of 411 individuals were 8.36 ± 6.22 mmHg for SBP and 5.69 ± 3.97 mmHg for DBP [25]. In the same year, Qin et al. proposed a model based on the ResNet34 framework and a channel attention mechanism. On the MIMIC-II database, the predicted MAE $\pm$ SD for SBP and DBP were 5.98 ± 7.79 mmHg and 3.24 ± 4.94 mmHg [26]. These advancements signify the ongoing innovation and development of deep learning in BP estimation, particularly in integrating diverse network structures and mechanisms to capture both temporal and spatial information. However, another

problem worth noting is that the above studies have only tested the performance of their models on one type of pulse wave. There has not yet been a BP model that has been verified to be universally applicable to multiple pulse waves.

The purpose of this paper is to develop a general pulse wave deep learning (PWDL) approach to tackle the limited accuracy and generalizability of current cuffless BP estimation. In this study, we collected three types of pulse wave data: PPG, iPPG, and PPWr, and proposed a novel hybrid deep BP model OVAR-BPnet for improved feature learning and enhanced generalizability. We first construct population datasets and employ data preprocessing to boost the quality and uniformity of the signal. Subsequently, we deploy an omni-scale convolution (OC) subnet and a Vision Transformer (ViT) subnet in the single-period and multi-period waveform feature extraction units, respectively, to extract waveform features. In the blood pressure regression unit, the model fuses the feature from the preceding subnet with phenotypic features, producing estimated BP values through a Multi-layer Perceptron (MLP) subnet. Building upon the previous feature extraction and BP regression units [27], our new model integrates data augmentation and label sequence regularization strategies to further improve the feature learning ability. A variety of performance verifications on private and public datasets and model interpretability research were studied. Notably, this is the first study to assess BP estimation performance concurrently across three types of pulse waves.

The organization of this paper is as follows: Section II introduces the three datasets and describes the architecture of the OVAR-BPnet model. Section III validates the model performance using three types of pulse wave datasets and conducts an interpretability study. Section IV provides an in-depth discussion and Section V concludes with a summary.

II. METHODOLOGY

As shown in Fig. 1, the proposed PWDL approach for the cuffless BP estimation includes three stages: data collection, data preprocessing, and hybrid deep BP model development. This section will provide the implementation details.

A. Data Collection

Leveraging two proprietary computer-aided medical diagnostic systems—the pulse diagnostic apparatus and the facial diagnostic apparatus—our study successfully collected two private datasets (PPWr and iPPG) [25], [27], as depicted in Fig. 1. The pulse diagnostic apparatus accurately positions three-finger contacts on the Cun, Guan, and Chi of the subject's radial artery through vision and robotic arm movement [28]. A force controller increases the pressure from 50 g to 250 g with a gradient of 20 g, allowing flexible film pressure sensors to capture subcutaneous radial artery pressure signals generated by blood pressure changes stably. Each pressure signal has 500 sampling points in 5 seconds, and the pressure signal from Guan with the highest signal-to-noise ratio is finally selected to represent the original PPWr signal. The facial diagnostic apparatus collects iPPG by positioning the subject's face in the central lighting chamber and using 530 ± 5 nm green light to illuminate the subject's face. Next, the apparatus captures a set of 8s facial

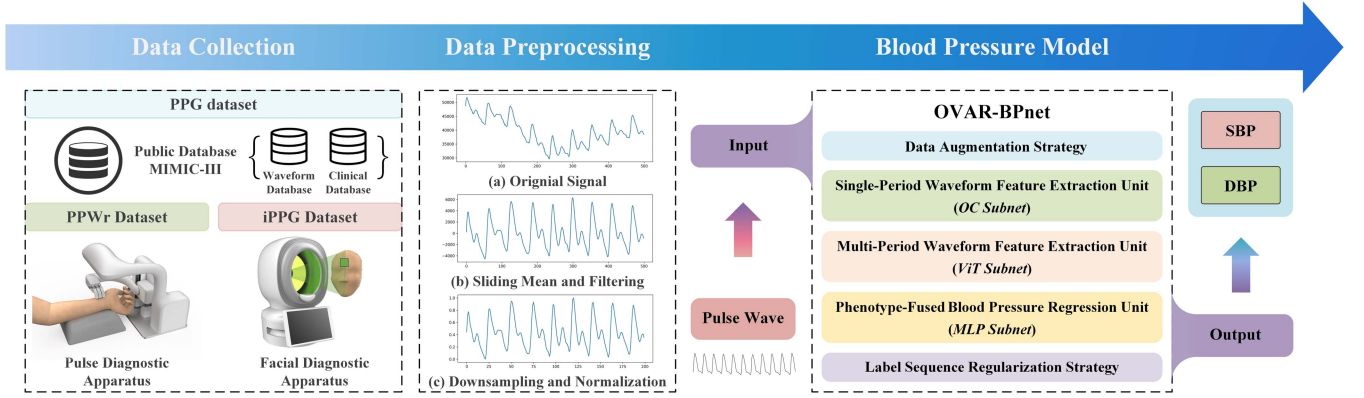


Fig. 1. Overall pipeline diagram of the PWDL approach for the cuffless blood pressure estimation method, from data collection to the blood pressure model OVAR-BPnet based on deep learning and pulse wave learning.

images at a frame rate of 40 frames/s. By extracting the average pixels of the regions of interest in these images, the original iPPG signal is finally obtained [29].

The PPWr and iPPG datasets comprise 2073 and 411 participants, respectively. Each subject contributes a set of pulse wave signals, BP labels measured by an electronic sphygmomanometer, and phenotypic information [25], [27]. The BP labels follow a normal distribution, the SBP ranges from 80 to 167 mmHg and DBP from 40 to 97 mmHg in the PPWr dataset. The iPPG dataset has an SBP range of 80 to 139 mmHg and a DBP range of 42 to 87 mmHg. The phenotypic information included gender, age, height, weight, and BMI. This research was approved by the Institutional Review Board and the Ethics Committee of Fudan University, with informed consent obtained from all subjects.

Furthermore, to make a fair comparison between this study and existing studies, we used the Medical Information Mart for Intensive Care III (MIMIC-III) waveform database and the clinical database in the collection of PPG dataset. The waveform database contains 22317 long-term waveform records from 10282 different ICU patients [30]. The records include digitized signals such as ECG, ABP, and PPG, with a sampling frequency of 125 Hz. In addition, the clinical database provides their phenotypic information, diagnosis, and drug treatment records. We successfully extracted fingertip PPG and beat-to-beat ABP records of 520 patients and obtained their phenotypic information (age and gender) from the MIMIC-III database. Fig. 2 illustrates the age and BP distribution using kernel density estimation curves and histograms, while a pie chart depicts the gender distribution. The Shapiro-Wilk statistics of age and BP are close to 1, indicating that they obey the normal distribution. In terms of gender, there are 299 males and 221 females. Typically, we limit the range of SBP to 80 to 180 mmHg and the range of DBP to 40 to 110 mmHg [31]. The dataset contains 503425 PPG signal fragments, with almost the same number of fragments for each patient.

B. Data Preprocessing

The PPWr dataset and the iPPG dataset undergo identical preprocessing protocols. The data preprocessing in Fig. 1 depicts the transformation of PPWr signals. The steps of this preprocessing include sliding mean, filtering, downsampling,

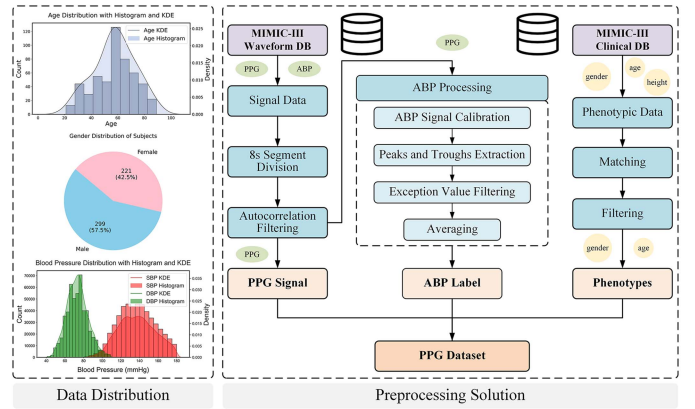


Fig. 2. PPG dataset's data distribution and preprocessing solution.

and normalization in sequence. Pulse wave signals frequently exhibit significant baseline drift due to bodily movements, which the sliding mean method effectively mitigates. Subsequently, we applied a fifth-order Butterworth filter with a bandpass range from 0.5 Hz to 10 Hz, adeptly isolating essential signal characteristics while excluding irrelevant frequency components [32]. Additionally, we carried out a downsampling procedure to standardize the sampling frequency of all pulse wave signals to 40Hz. The final step involves normalization, which not only enhances the training efficiency and generalizability of the model but also shields the model from the effects of waveform amplitude variations.

Fig. 2 also demonstrates the preprocessing solution for the PPG dataset. Long-term PPG and ABP signals are uniformly segmented into 8-second windows, each containing 1000 data points. An autocorrelation filter, with a threshold of 0.75, is then applied to exclude segments with corruption or poor waveform periodicity [33].

$$R(\tau) = \frac{1}{N} \sum_{t=0}^N \xi(t) \cdot \xi(t-\tau) \quad (1)$$

where $R(\tau)$ represents the autocorrelation coefficient when the offset time is τ , $\xi(t)$ is the original signal, $\xi(t-\tau)$ is the offset signal, and N is the number of overlapping signal points.

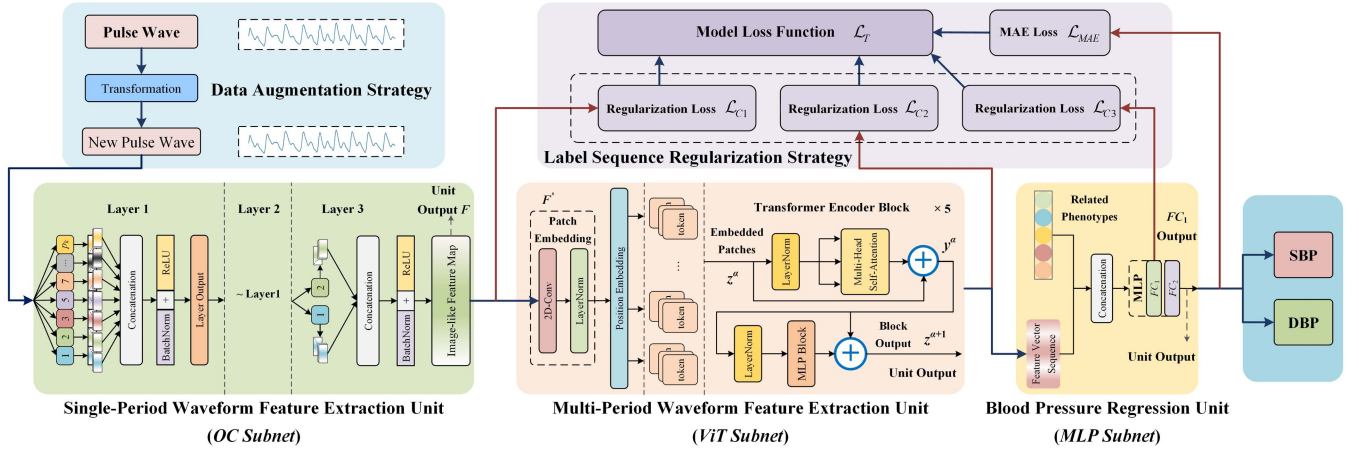


Fig. 3. Architecture of OVAR-BPnet model with single-period waveform feature extraction unit, multi-period waveform feature extraction unit, phenotype-fused blood pressure regression unit, data augmentation strategy, and label sequence regularization strategy.

The subsequent preprocessing process of the PPG signal segments is the same as that of the other two pulse waves. The ABP signal segments filtered by the autocorrelation filter still needs to be processed to obtain the BP label of the 8s segments. These processing steps include signal calibration, peaks and troughs extraction, as well as exception value filtering and averaging. The average of the peaks of a certain ABP signal segment represents the SBP label of the segment, while the average value of the troughs represents the DBP label.

C. Model Architecture

The design of the BP model OVAR-BPnet proposed in this paper is a novel hybrid deep learning model and is based on the PWA. The input is a pulse wave signal with a shape of $1 \times L$ (320 or 200), and then the model automatically extracts the single-period and multi-period waveform features of the pulse wave signal, and fuses BP-related phenotypes to output a shape of 1×2 estimates of blood pressure (SBP and DBP).

Fig. 3 unveils the OVAR-BPnet model's architecture. As the pulse wave is fed into the network, a data augment strategy initially reconstitutes it into an updated waveform. Then it is sent to the single-period and the multi-period waveform feature extraction units in sequence to mine waveform features. The essence of the first unit is an OC subnet, devised to help the model extract the best local features of the single-period waveform derived. The subsequent unit, structured around the Vision Transformer architecture, employs a ViT subnet tailored to apprehend the comprehensive interrelations amongst multiple waveforms. In the final unit, the model fuses the feature sequence with phenotypic information and engages a MLP subnet to deduce the estimated SBP and DBP. Additionally, we also design a regularization strategy to improve the model's pulse wave learning ability. In what follows, we will elaborate on the work of each unit and strategy.

D. Model Design

1) *Single-Period Waveform Feature Extraction Unit*: Certain types of pulse waves show unique consistency within the same

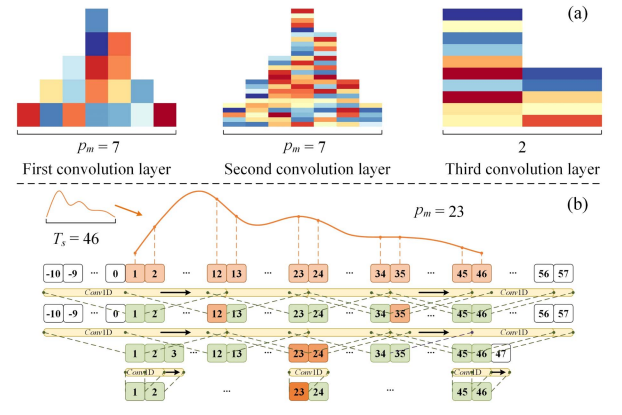


Fig. 4. Simplified structure of the three convolution layers and the core convolution process in the OC subnet.

subject, showing nearly identical repeating patterns over time, but with slight variations. Different types of pulse waves in the same subject have different morphologies and almost no difference in periodicity. The same type of pulse wave in different subjects has significant differences in periodicity and morphology due to individual differences. Therefore, to capture the best local features of pulse waves of different subjects, this paper uses an omni-scale CNN in the single-period waveform feature extraction unit [34]. It is designed based on Goldbach's conjecture and can capture the best receptive field (RF) of the time series, thereby achieving the best performance of one-dimensional (1D) CNN.

In order to represent the structure of the three convolutional layers more clearly in the OC subnet, we present a simplified case using a heatmap in Fig. 4(a). The following equation indicates that the kernel sizes for the first two layers consist of prime numbers ranging from 1 to p_m , while the final layer's kernel sizes are fixed at 1 and 2. To satisfy this convolution method, each layer employs a zero mask to generate multiple convolution kernels with varying sizes. The heatmap illustrates a scenario where the maximum kernel size, p_m , is 7, with the number of channels (input, output) for the first two convolution layers being

(1, 5) and (5, 5), respectively, and (5, 2) for the final layer. The output channels for each convolution layer can be independently configured, with all kernels within a layer operating in parallel. The output of the second convolution layer is 1024 channels, and the final layer is set to 512 channels, enhancing the deep convolution and information compression capabilities of the OC subnet. As depicted in Fig. 3, the output of the convolutional layers is produced by splicing feature maps, followed by batch normalization and ReLU activation.

$$P(i) = \begin{cases} \{1, 2, 3, 5, 7, \dots, p_m\}, & i \in \{1, 2\} \\ \{1, 2\}, & i = 3 \end{cases} \quad (2)$$

To visually demonstrate the process by which the OC subnet extracts features from a single period waveform, we illustrate the core process of the three-layer convolution in Fig. 4(b). We analyzed the pulse wave period (T_s) of 520 subjects and found that T_s usually does not exceed 46 sampling points at 40 Hz. Therefore, we design the convolution kernels with a stride of 1 and apply bilateral zero-padding to ensure that the shape of the convolution output is consistent with the input pulse wave data. Based on the characteristics of this convolution method, we set the convolution kernel size to a prime number slightly beyond the half-period length ($p_m = 23$). After the three layers of convolution, the RF size can cover a complete pulse wave period length, thereby extracting the features of a single period waveform. Additionally, the first two convolution layers use prime-sized kernels across the full range, ensuring the model's OC capability and adaptively capturing the optimal RF and local features of a single-period waveform.

2) Multi-Period Waveform Feature Extraction Unit: This unit aims to enable the BP model to focus on the interdependencies between multiple waveforms from a time series perspective. Traditional recurrent neural networks (RNNs), including Gated Recurrent Unit (GRU) and LSTM, are challenging to apply to biomedical signals like pulse waves. Additionally, this unit requires the capability to process the two-dimensional (2D) feature map output by the OC subnet. Consequently, this study introduces the ViT architecture [35]. Although the ViT was originally tailored for image recognition, its self-attention mechanism is well-suited for analyzing time series data, making it an ideal choice for a subnet in the model.

In this unit, to adapt the 2D feature map output by the OC subnet $F \in \mathbb{R}^{bs \times C \times W}$ to the input requirements of the ViT subnet, we expand the dimension by adding a dimension at index 1, transforming it into a single-channel feature map $F^* \in \mathbb{R}^{bs \times 1 \times H \times W}$. Here, $H = C$, and bs = batch size. Then follows the Patch Embedding. The ViT subnet uses 2D convolution to divide the feature map F^* into n patches (x) with a stride size of 2, mapping each into a 40-length embedding vector ($token$) and applying layer normalization (LN). Through these operations, the ViT subnet transforms features corresponding to multi-period pulse waves into standardized tokens. Since the ViT subnet lacks inherent sequential processing capabilities like RNN, Position Embedding needs to be added to each token to provide temporal sequence information (p). This enables the ViT subnet to understand the temporal order of multiple

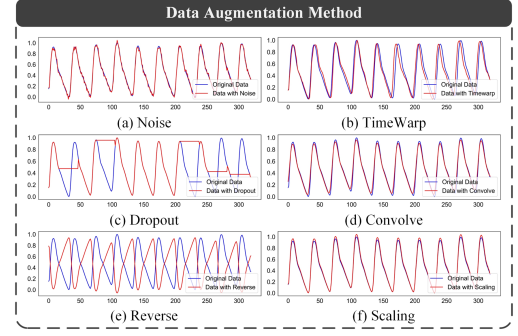


Fig. 5. Pulse waveform variations via data augmentation methods.

tokens, which is crucial for capturing the temporal correlation and sequential patterns among multiple period waveforms. The Patch Embedding and Position Embedding are as follows.

$$token_i = LN(Conv2D(x_i)), i = (1, \dots, n) \quad (3)$$

$$z = [token_1, token_2, \dots, token_n] + p \quad (4)$$

As shown in Fig. 3, the transformer encoder block (TEB) is located at the end of the ViT subnet, and the number of TEBs affects the depth of the model. It is the core component of the ViT architecture, including the Multi-Head Self-Attention (MHSA) mechanism, enabling the model to understand the temporal dependencies and between multiple period waveforms. Each TEB contains an LN , a $MHSA$, a MLP block, and two residual connections. The computation of TEB is as follows.

$$y^\alpha = MHSA(LN(z^\alpha)) + z^\alpha \quad (5)$$

$$z^{\alpha+1} = MLP(LN(y^\alpha)) + y^\alpha \quad (6)$$

where z^α is the input of the α -th TEB, y^α is the result of the first residual connection, $z^{\alpha+1}$ is the result of the second.

3) Phenotype-Fused Blood Pressure Regression Unit: To enhance the precision and customization of BP estimation, we devised a blood pressure regression unit integrating phenotypic data at the model's end. Consciously omitting the classification head within the ViT architecture, our aim is to harness the TEB for capturing global features from pulse wave time series, acknowledging BP estimation as a regression challenge. Crucially, the normalized phenotype data merges with the feature vector sequence output by the ViT subnet, rectifying the constraints of single-channel pulse wave signal characteristics and furnishing personalized information for bolstering estimation accuracy. In the model's ultimate unit, we employ an MLP subnet comprising two fully connected layers (FC_1 and FC_2) to execute BP regression.

4) Data Augmentation Strategy: In the field of time series tasks, data augmentation strategies play an important role in increasing model robustness and model generalization. Therefore, this study also tried and adopted a series of data augmentation methods, including noise addition (Noise), time warp (Timewarp), data loss (Dropout), convolution (Convo), signal flipping (Reverse), and signal scaling (Scaling). Fig. 5 shows the variations of the PPG waveform after using the six data augmentation methods.

It is important to highlight that while these data augmentation techniques have been extensively proven effective in the realm of brainwave (EEG) classification [36], their application in the demanding task of pulse wave-based BP estimation represents a pioneering exploration with significant research and practical potential. Given the intricate nature and substantial interindividual variability of physiological signals, such as the pulse wave, challenges persist in effectively utilizing physiological information detection. Introducing greater diversity in data sources may offer a viable solution [37]. The investigation into data augmentation strategies in this study is grounded in *tsaug*, a Python package tailored for enhancing time series data. Our objective is to implement select data augmentation techniques mentioned above to enrich the diversity and complexity of training samples, a pivotal step in enhancing the accuracy, generalization capability, and robustness of the BP model.

5) Label Sequence Regularization Strategy: Furthermore, this study closely combines features learned from pulse waves with their downstream BP estimation task, proposing a regularization strategy called neighborhood feature continuity loss. This strategy is based on a reasonable assumption: as a physiological signal, samples with adjacent BP labels in the pulse wave signal should exhibit higher similarity in feature performance. Therefore, the strategy aims to constrain the features output by each subnet in the model to ensure that the Euclidean distance in the feature space between samples with adjacent labels is minimized.

In each batch, we specifically analyze pairs of training samples with adjacent labels (y_i, y_{i+1}), computing their respective features ($X_{y_i}, X_{y_{i+1}}$). The average Euclidean distance between these feature pairs is calculated accordingly. Notably, for the BP regression unit involving fused phenotypes, we exclusively evaluate the output features of the FC_1 layer. It is common within a batch to encounter samples lacking adjacent labels. For such cases, the average Euclidean distance from other samples is not computed.

We define the continuity losses for domain features generated by the single-period waveform feature extraction unit, multi-period waveform feature extraction unit, and phenotype-fused BP regression unit as $\mathcal{L}_{C1}, \mathcal{L}_{C2}$, and $\mathcal{L}_{C3}$, respectively. The continuity loss equation for any neighborhood feature, $\mathcal{L}_C$, is delineated as follows.

$$\mathcal{L}_C = \sum_{i=1}^{n-1} I[y_{i+1} - y_i = 1] \times \Phi_{dist}(X_{y_i}, X_{y_{i+1}}) \quad (7)$$

within the equation, I represents the indicator function, which is 1 when the values of the pair of labels are adjacent, otherwise it is 0. Φ_{dist} represents the average Euclidean distance. The average Euclidean distance is as follows.

$$\Phi_{dist}(A, B) = \frac{1}{|A| \cdot |B|} \sum_{a \in A} \sum_{b \in B} \|a - b\|_2 \quad (8)$$

here, A and B represent two sets, $|\cdot|$ represents the number of elements in the set, and $\|a - b\|_2$ represents the Euclidean distance between element a and element b .

E. Experimental Setup

The experimental setups used in this research for BP estimation experiments on three types of datasets (PPG, iPPG, and PPWr) are as follows. The same setups are that the model is trained in an Ubuntu 23.10 environment containing Python 3.9.18, PyTorch 1.12.0, CUDA11.6, and cuDNN8302, and the GPU is an NVIDIA RTX 4090. This paper uses ten-fold cross-validation to ensure the robustness of the verification results. The ratio of the training set and the test set is 9:1, and the mean value of the cross-validation is used as the result. The learning rate during model training is $5e-3$, the weight decay coefficient is $1e-4$, the batch size is 32, the optimizer is SGD, and an early stopping mechanism ends without performance improvement beyond 50 epochs. The evaluation indicators of BP estimation accuracy are $MAE \pm SD$ and mean error (ME) $\pm SD$, and the evaluation criteria of model performance come from the British Hypertension Society (BHS) and the Association for the Advancement of Medical Instrumentation (AAMI) [38]. The OVAR-BPnet model includes a regression loss $\mathcal{L}_{MAE}$ and three neighborhood feature losses $\mathcal{L}_{C1}, \mathcal{L}_{C2}$, and $\mathcal{L}_{C3}$. The overall loss function $\mathcal{L}_T$ as follows.

$$\mathcal{L}_T = \mathcal{L}_{MAE} + \omega_1 \mathcal{L}_{C1} + \omega_2 \mathcal{L}_{C2} + \omega_3 \mathcal{L}_{C3} \quad (9)$$

among them, ω_1, ω_2 , and ω_3 are the weights of $\mathcal{L}_{C1}, \mathcal{L}_{C2}$, and $\mathcal{L}_{C3}$ respectively, which play a role in regulating the importance of the loss component.

The different experimental setups involve the methods of cross-validation and the sizes of the datasets. For the PPG dataset, cross-validation is usually segregated into subject-wise and sample-wise methods. The subject-wise method partitions the dataset according to individual subjects, ensuring no overlap between the subjects in the training and testing sets. This method yields a dataset known as the PPG-SuW dataset. On the other hand, the sample-wise method partitions the training and testing sets based on numerous samples from each subject, yielding the PPG-SaW dataset. In addition, the PPG-SaW method requires sufficient samples for each subject. Many researchers utilizing public datasets opt for one of these partitioning methods. This paper employs both methods to facilitate a more comprehensive comparison of model performance. As for the iPPG and PPWr datasets, the number of labeled samples for each subject is only 1, so the division of the data sets is the same as that of PPG-SuW. In terms of dataset size, both the PPG-SuW and PPG-SaW datasets have 520 subjects and 503425 samples. The number of subjects in the iPPG and PPWr datasets is also the number of their samples, which are 2073 and 411 respectively. Our model will undergo performance verification on four datasets in experiments.

III. RESULTS

A. Tuning of Hyperparameters

The hyperparameters that need to be accurately adjusted in the OVAR-BPnet model include data augmentation methods, model structure parameters, and loss function weights, which are crucial to the optimization of model performance. We first

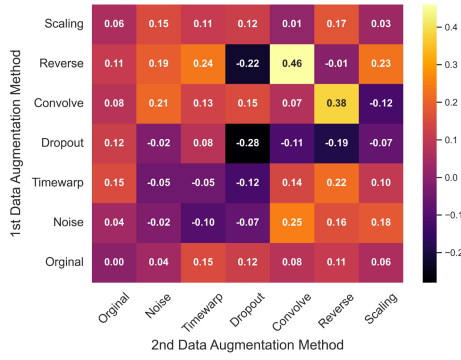


Fig. 6. Effects of data augmentation methods visualized in a heatmap.

studied data augmentation strategies, but only selected some combinations of data augmentation methods to ensure a balance between efficiency and effectiveness. Fig. 6 shows the comprehensive gain impact of different data augmentation methods and their combinations on the accuracy of SBP and DBP estimation in the form of a heatmap. Specifically, we implemented 30% random online sampling data augmentation for each batch of samples and used at most two data augmentation methods. This not only improves the diversity of waveform morphology, but also retains the characteristics of the original waveform. The vertical and horizontal axes of the heatmap mark the first and second data augmentation methods, and the positive values in the grid indicate the improvement in model performance compared to when the data augmentation strategy was not used. Through analysis, this paper determines a combination of flipping and convolution as data augmentation methods in the model architecture, which can help improve the overall accuracy of BP estimation by 0.46mmHg.

The structure parameters are particularly essential, mainly including the maximum convolution kernel size (p_m) in the OC subnet and the number of TEBs in the ViT subnet. Based on the recommendation of the previous research, we set the value of p_m to 23 to obtain the optimal RF for capturing a single-period waveform [27]. For the number of TEBs, the study increased it from 1 to 10, and established that setting 5 TEBs yields the most favorable BP estimation performance. Although more TEBs achieved similar results, an increase in model depth results in higher memory and time. Regarding the configuration of the loss function weights ($\omega_1, \omega_2, \omega_3$), we tested various combinations by reducing the values from $1e-1$ to $1e-4$. Remarkably, when $(\omega_1, \omega_2, \omega_3)$ is $(1e-3, 1e-4, 1e-3)$, the MAEs of SBP and DBP of the OVAR-BPnet model are minimized.

B. Model Performance Evaluation on Four Datasets

Utilizing hyperparameters identified in the previous subsection, this study embarks on BP estimation experiments utilizing three distinct pulse wave datasets: PPG, iPPG, and PPWr. These experiments aim to evaluate the accuracy and generalizability of the OVAR-BPnet model across multiple types of pulse waves. This marks the first time that a study has performed simultaneous

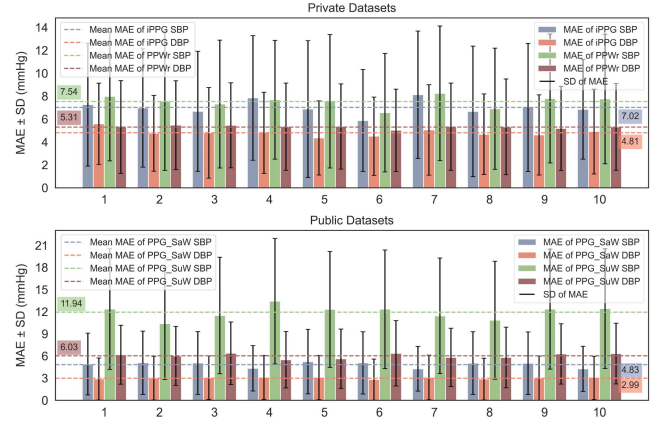


Fig. 7. Ten-fold cross-validation results of four datasets.

BP estimation experiments on three types of pulse waves, providing a comprehensive assessment of the model's generalizability in pulse waves. The analysis inputs four previously divided pulse wave datasets (iPPG, PPWr, PPG-SuW, and PPG-SaW) into the model. Excitingly, as shown in Fig. 7 and Table I, the BP estimation of the OVAR-BPnet model on four datasets exhibits the desired performance. Fig. 7 delineates the MAE and the SD for each fold within each dataset, alongside the aggregate MAE mean. Employing $MAE \pm SD$ as the indicator of accuracy, the model exhibits commendable BP estimation capabilities on the iPPG dataset, with SBP accuracy at 7.02 ± 5.28 mmHg and DBP accuracy at 4.81 ± 3.57 mmHg. Although BP studies on the PPWr dataset are scarce, our findings suggest significant potential for BP estimation using this pulse wave, with our model estimating SBP and DBP at 7.54 ± 5.59 mmHg and 5.31 ± 3.82 mmHg, respectively. The PPG-SuW dataset, categorized by subject, indicates that substantial individual variances within public dataset may adversely affect outcomes, evidenced by estimated $MAE \pm SD$ for SBP and DBP at 11.94 ± 8.02 mmHg and 6.03 ± 4.08 mmHg, respectively. Conversely, the PPG-SaW dataset, categorized by sample, ensures that both training and test sets include data from the same subjects, yielding more favorable model estimation results, with SBP and DBP $MAE \pm SD$ at 4.83 ± 3.90 mmHg and 2.99 ± 2.93 mmHg, respectively.

Furthermore, this paper leverages internationally recognized BP measurement standards, namely BHS and AAMI, to evaluate the model's accuracy. According to the BHS standard grade classification details and experimental results in Table I, the OVAR-BPnet model's estimation of SBP and DBP on the iPPG dataset reached BHS grade B. On the PPWr dataset, the model's estimation of SBP and DBP reached grade C and grade B respectively. In the PPG dataset by subject, the BHS grades for SBP and DBP were D and C, respectively. In the PPG dataset classified by samples, the BHS grades of SBP and DBP are better than the A grade. The BP measurement standards defined by AAMI are in Table I. The AAMI standard requires the number of testers to be more than 85 people. The PPWr dataset meets this requirement, while the PPG and iPPG datasets can meet this requirement through cross-validation. The results show that the

TABLE I
COMPARISON WITH THE BHS STANDARD AND THE AAMI STANDARD

		≤5 mmHg	≤10 mmHg	≤15 mmHg	Grade			ME (mmHg)	SD (mmHg)	Subjects	AAMI Result
BHS		60%	85%	95%	A	AAMI		≤ 5	≤ 8	≥ 85	
		50%	75%	90%	B						
		40%	65%	85%	C						
iPPG	SBP	53.5%	80.4%	92.1%	B	iPPG	SBP	-0.67	8.61	411	Fail
	DBP	56.4%	88.3%	98.6%	B		DBP	-0.23	5.82		Pass
PPWr	SBP	40.2%	71.2%	88.3%	C	PPWr	SBP	0.08	9.29	2073	Fail
	DBP	53.1%	86.4%	98.8%	B		DBP	-0.58	6.48		Pass
PPG-SuW	SBP	25.1%	46.9%	65.5%	D	PPG-SuW	SBP	-0.27	14.22	520	Fail
	DBP	46.6%	80.6%	98.0%	C		DBP	0.08	7.24		Pass
PPG-SaW	SBP	61.3%	88.8%	97.3%	A	PPG-SaW	SBP	-0.15	6.17	520	Pass
	DBP	82.3%	96.3%	99.2%	A		DBP	-0.21	4.16		Pass

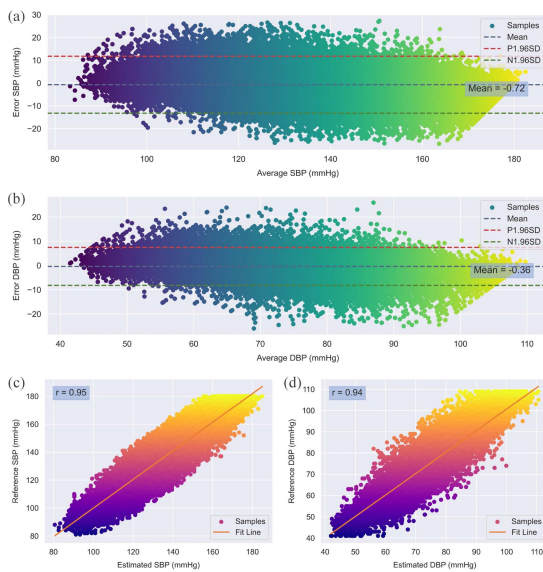


Fig. 8. Consistency analysis and correlation analysis of PPG dataset results. (a) And (b) are the Bland-Altman plots of SBP and DBP. (c) And (d) are the Pearson correlation plots of SBP and DBP.

estimation errors of SBP and DBP by the OVAR-BPnet model on the PPG-SaW dataset fully meet the requirements of the AAMI standard. However, in the iPPG, PPWr, and PPG-SuW datasets, the model only meets the requirements of ME on SBP and does not meet the requirements on SD, but it meets the requirements on DBP.

This study further performed consistency and correlation analyses between the estimated and reference BP values derived from the model across four datasets. Owing to space constraints, Fig. 8 exclusively presents the BP estimation analysis results for the model using a sample-wise partitioning on PPG data. The depicted data originate from samples within a specific fold of the PPG-SaW test dataset. Fig. 8(a) and (b) illustrate the consistency analysis for SBP and DBP, utilizing the Bland-Altman method. The ME $\pm$ SD for SBP and DBP was recorded at -0.72 ± 6.37 mmHg and -0.36 ± 3.99 mmHg, respectively. Most data points in figure fall within the limits of agreement range (ME $\pm 1.96 \times$ SD), demonstrating the model's substantial consistency

in matching the BP estimates with the reference values from the PPG dataset. Regarding correlation analysis, this paper employs the Pearson correlation coefficient, as depicted in Fig. 8(c) and (d). The correlation coefficients (r) for the estimated and reference values of SBP and DBP from the PPG dataset stand at 0.95 and 0.94, respectively, indicating a strong positive linear correlation between them.

C. The Importance of Three Subnets and Two Strategies

The OVAR-BPnet model's architecture underwent validation via ablation experiments conducted on the PPG dataset in sample-wise method, as detailed in Table II. In these experiments, we tweaked the model's architecture while consistently preserving the MLP subnet dedicated to regression tasks. The results of Table II demonstrate that the synergy of the OC subnet and ViT subnet extracts pulse wave waveform characteristics more effectively than a single subnet. The combination of OC subnet and MLP subnet can achieve good BP estimation results, because the design intention of OC subnet is to process temporal data. However, the model performance of the ViT subnet combined with the MLP subnet is not ideal, which may be due to the fact that 1D pulse wave sequences do not possess the features of image data even after being upscaled into single channel data ($1 \times 1 \times L$). Therefore, ViT cannot fully capture the global dependencies in the pseudo image data through its *MHSA* mechanism.

Incorporating phenotypic information as features enhances the OVAR-BPnet model's performance on the PPG-SaW dataset. However, this benefit was not observed in additional experiments on the PPG-SuW dataset. This highlights that such phenotypic information may provide personalized insights not obtainable from pulse waves alone but is limited by individual variability across subjects. The inclusion of data augmentation strategies further marginally enhances the model's BP estimation capabilities. Finally, the addition of a neighborhood feature continuity loss culminates in the OVAR-BPnet model achieving optimal SBP and DBP estimation accuracy. Therefore, the single-period and multi-period waveform feature extraction units, phenotype-fused blood pressure regression unit, data augmentation strategy, and regularization strategy are all key

TABLE II
RESULTS OF ABLATION EXPERIMENTS ON THE OPTIMAL MODEL STRUCTURE

OC Subnet	ViT Subnet	Phenotypic info.	Data-Aug	L_C	SBP (mmHg)		DBP (mmHg)	
					MAE	SD	MAE	SD
✓					6.29	5.87	4.72	4.41
	✓				16.14	11.20	9.65	6.94
✓	✓				5.64	5.36	3.67	3.74
✓		✓			6.18	5.93	4.35	4.08
✓	✓	✓			5.36	4.90	3.41	3.54
✓	✓	✓	✓		5.05	4.26	3.18	3.42
✓	✓	✓	✓	✓	4.83	3.90	2.99	2.93

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT MODELS BASED ON PRIVATE DATASETS

Dataset	Model	MAE $\pm$ SD (mmHg)	
		SBP	DBP
iPPG	CNN [14]	11.78 $\pm$ 7.97	7.79 $\pm$ 5.93
	CNN-LSTM [39]	10.17 $\pm$ 7.90	6.97 $\pm$ 4.95
	Hybrid-D1DCnet [25]	8.36 $\pm$ 6.22	5.69 $\pm$ 3.97
	OVAR-BPnet (This Work)	7.02 $\pm$ 5.28	4.81 $\pm$ 3.57
PPW _r	CNN [14]	12.08 $\pm$ 9.42	8.41 $\pm$ 6.08
	CNN-LSTM [39]	10.98 $\pm$ 7.50	8.25 $\pm$ 5.25
	OCViTnet [27]	8.11 $\pm$ 6.60	5.92 $\pm$ 4.86
	OVAR-BPnet (This Work)	7.54 $\pm$ 5.59	5.31 $\pm$ 3.82

components of this study. They are indispensable in contributing to the model's performance. Compared to other incomplete architectures, the OVAR-BPnet model shows superior BP estimation performance.

D. Performance Comparison With Other Works

To clearly demonstrate the superior performance of the OVAR-BPnet model in estimating BP across both private and public datasets, this study employed different validation strategies. For BP estimation experiments based on private datasets (iPPG and PPW_r), this paper established a series of end-to-end neural network models to achieve performance competition among different model architectures in Table III. These models include CNN [14], CNN-LSTM (PPNet) [39], Hybrid-D1DCnet [25] and OCViTnet [27]. The results showed that the CNN model has the worst results, the hybrid structure CNN-LSTM model is not as effective as Hybrid-D1DCnet and OCViTnet, and OVAR-BPnet has the highest BP estimation results. This is a reasonable result, because previous research has proven that OCViTnet is better than Hybrid-D1DCnet, and the more powerful OVAR-BPnet proposed in this paper is an upgrade to OCViTnet. The data augmentation strategy and the label sequence regularization strategy studied in this paper can provide greater performance improvement for the OCViTnet model with a hybrid structure. In addition, the experimental results also illustrate that the data augmentation strategy and the regularization strategy of the model are equally effective on the other two types of pulse wave.

Validation of model performance on the public dataset was usually compared with the results of studies that also perform cuffless BP estimation on public datasets. Table IV lists the BP

estimation studies using the MIMIC-II, UCI, and MIMIC-III databases in the past five years. These studies all use the PPG signal as the only signal source of the BP model [19], [24], [26], [40], [41], [42], [43]. It should be noted that since different ways of dividing the dataset will lead to large differences in model accuracy, this paper further subdivided these studies into the subject-wise group and the sample-wise group. In the subject-wise group, the MAE of SBP estimated by the OVAR-BPnet model is the lowest. However, the MAE on DBP was only slightly inferior to Samimi's study. The reason may be that their dataset size is larger than ours [43]. The results for the sample-wise group aligned with our expectations, demonstrating that our model outperforms others. This underscores the model's potential in the field of cuffless BP estimation.

It is worth noting that the studies [24], [26], [40], [41], despite employing hybrid structured models such as Seq2Seq-Attention, ResNet-GRU, ResNet-LSTM, and MSA-ResNet, fail to be comparable to the final performance of the OVAR-BPnet model. The main reason is that the single-period and multi-period waveform feature extraction units enable the model to fully capture the local features of a single period of the pulse wave and the interdependence between multiple pulse wave periods. In addition, the deployment of data augmentation and regularization strategies can enrich data and improve the feature learning capabilities of the model, respectively.

E. Interpretability Analysis of Pulse Wave Learning

Deep learning technology is usually considered a black box operation, and it is not easy to know how the BP model learns BP information from specific pulse wave signals [44]. In order to conduct interpretability research in this direction, we used t-distributed stochastic neighbor embedding (t-SNE) to visualize the feature transformation of a set of PPG test data of a subject in the model. t-SNE technique reveals the space structure of data by simulating the similarities between high-dimensional data points and mapping them to a 2D plane. After multiple experiments, we optimized the hyperparameters of t-SNE, setting the perplexity to 20, early exaggeration to 12.0, iteration number to 1000, and keeping the learning rate and random seed at default values. Fig. 9 shows the distribution patterns and transformations of the PPG signals and their features generated after processing through the subnets of the OVAR-BPnet model, which are derived from the test set data of three subjects (A, B, and C).

TABLE IV
PERFORMANCE SUMMARY OF BP ESTIMATION METHODS USING PUBLIC DATABASE

Author (Year)	Signals	Dataset (cases)	Model	SBP (mmHg)				DBP (mmHg)			
				MAE	SD ¹	ME	SD ²	MAE	SD ¹	ME	SD ²
Subject-Wise											
Slapničar (2019) [41]	PPG	MIMIC-III (510)	ResNet-GRU	15.41	-	-	-	12.38	-	-	-
Aguirre (2021) [40]	PPG	MIMIC III (1131)	Seq2Seq-Attention	14.39	17.87	0.62	17.87	6.57	8.43	-0.49	8.43
Samimi (2022) [43]	PPG	MIMIC (590)	ANN	14.49	-	0.09	18.8	5.75	-	0.03	7.91
This work	PPG	MIMIC III (520)	OVAR-BPnet	11.94	8.02	-0.27	14.22	6.03	4.08	0.08	7.24
Sample-Wise											
Hajj (2021) [42]	PPG	MIMIC-II (942)	GRU	5.77	8.52	-1.24	10.21	3.33	5.02	-0.44	5.70
Wang (2022) [19]	PPG	UCI (304)	AlexNet	6.17	-	0.00	8.46	3.66	-	-0.04	5.36
Paviglianiti (2022) [24]	PPG	MIMIC (40)	ResNet-LSTM	7.12	-	-	-	3.53	-	-	-
Qin (2023) [26]	PPG	MIMIC-II (-)	MSA-ResNet	5.98	-	0.92	7.79	3.24	-	0.68	4.94
This work	PPG	MIMIC-III (520)	OVAR-BPnet	4.83	3.90	-0.15	6.17	2.99	2.93	-0.21	4.16

SD¹ is the standard deviation of MAE, and SD² is the standard deviation of ME.

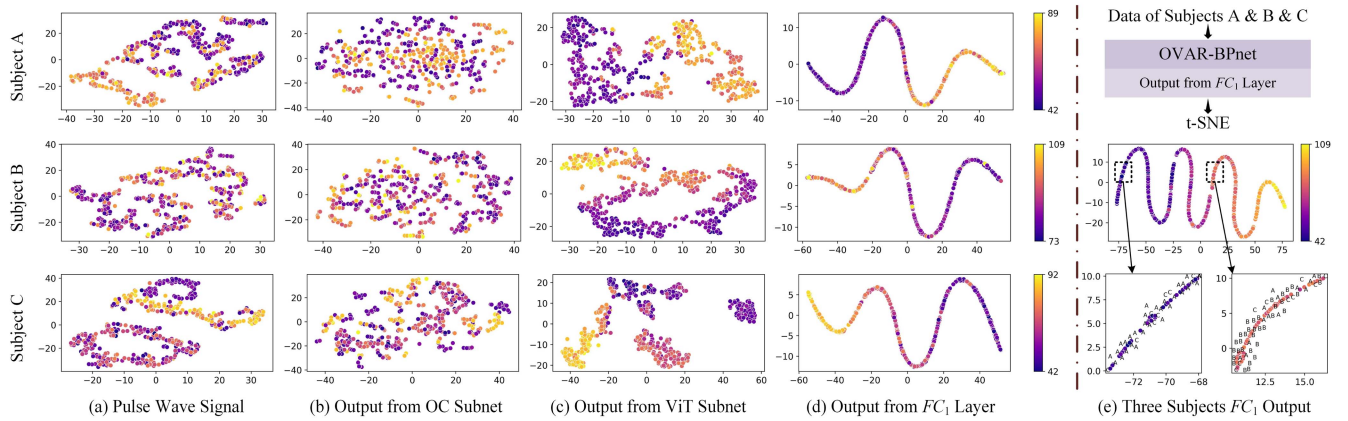


Fig. 9. Feature transformations in the PPG-based DBP estimation of subjects A, B, and C are illustrated using the t-SNE technique. (a)–(d) display the progression of individual feature distributions. (e) Shows the combined FC_1 outputs of all three subjects within the same plot, providing a zoomed-in view for observation. The color bar indicates the BP range.

Specifically, each point in these subplots represents an independent test sample of the subject, and the depth of the point's color indicates the magnitude of the corresponding BP values. It should be noted that due to the shuffling in data construction, there is no temporal continuity or correlation between the BP labels of these data points, but they are numerically continuous. Fig. 9(a)–(d) present t-SNE visualizations of individual subject data, including PPG signals, the 2D feature map output by the last convolution layer of the OC subnet, the feature vector output by the last TEB of the ViT subnet, and the feature vector output by the FC_1 layer of the MLP subnet. Fig. 9(e) displays the feature vector output by the FC_1 layer for data from three subjects combined.

In regression tasks, it is expected that the color of data points changes continuously with the spatial position [45]. The data points in Fig. 9(a) exhibit a certain degree of clustering, reflecting the similarity of the original pulse wave signal characteristics. Fig. 9(b) presents that the data points have more obvious clustering characteristics, indicating that the OC subnet extracts pulse wave features related to BP. The arrangement in Fig. 9(c) becomes even more pronounced, showcasing distinct structural coherence and depth. It indicates the ViT subnet's capability

to identify minute variations in BP across different pulse wave signals, a key aspect for precise BP estimations. Fig. 9(d) shows that the FC_1 layer of the final MLP subnet can make the feature distribution show clear continuity and structure, forming an almost perfect gradient curve. It is shown that the model's feature representation accurately reflects continuous changes in BP, which is ideal for BP regression tasks. Fig. 9(e) illustrates a zoomed-in view where points are labeled by subjects A, B, or C, indicating that similar BP intensities have closer features across different subjects. The interpretability study of the correlation between pulse waves and BP values remains underexplored. This paper clearly explains it through the t-SNE technique, which may become the focus of future research in the BP estimation field.

This study qualitatively compared the pulse wave learning capabilities of three baseline BP models: CNN [14], PPNet [39], Hybrid-D1DCnet [25], and our proposed OVAR-BPnet model using t-SNE. These baseline models were chosen to represent a range of architectures from simple to complex, specifically representing CNN, CNN-LSTM, and CNN-LSTM-Attention architectures, respectively. We unified their regression units with an FC_1 layer and then tested them on the same test set of data

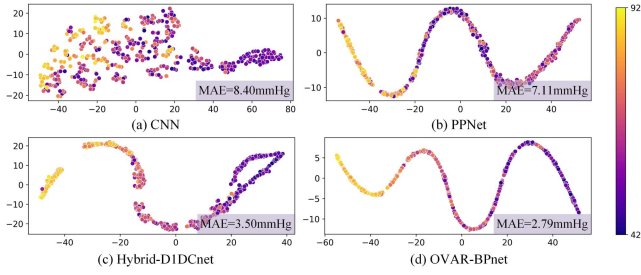


Fig. 10. Comparison of feature mapping results produced by FC_1 layer of regression units of different models using t-SNE technique.

from the same subject. Fig. 10 shows the 2D spatial mapping of their FC_1 layer features.

The results show that the lower the MAE, the stronger the consistency between the color gradient brought by the feature map and the continuous changes in BP values. The high MAE of the CNN model makes its features not show a curve structure. The MAE of the PPNet model is reduced to 7.11 mmHg. Its features show an initial curve structure, but the boundaries between BP categories are unreasonable. The Hybrid-DIDCnet model further reduces MAE by adding CNN layers and introducing a self-attention mechanism. Its features exhibit a more defined curvilinear structure, although some samples show insufficient learning. The OVAR-BPnet model proposed in this study performs the best, with an MAE of only 2.79 mmHg. Its feature visualization presents a continuous and clear BP gradient curve, proving its significant advantage in learning pulse wave features. This study demonstrates the effectiveness of t-SNE mapping as a qualitative assessment of BP model performance and provides a valuable tool for future research.

IV. DISCUSSION

A. Comprehensive Analysis of BP Estimation Results

According to the BP estimation accuracy of the two PPG datasets, the OVAR-BPnet model has achieved more competitive model estimation results on public datasets than existing research [19], [24], [26], [40], [41], [42], [43]. It is worth noting that the PPG-SuW, iPPG, and PPWr datasets all adopt subject-wise method, but the BP estimation performance of the model on them has not reached a high level like that of PPG-SaW. The reason is that the sample-wise method helps the model learn the prior knowledge of specific subjects during the training. This training method is similar to model calibration and fine-tuning. Therefore, it is easier to obtain an accurately estimated BP model by introducing subject-specific data into training.

Additionally, we found that although the experiments on the three pulse wave datasets all used the general PWDL approach, the OVAR-BPnet model's BP estimation performance on the iPPG and PPWr datasets exceeded the PPG-SuW dataset. This might be attributed to the fact that subjects in the two private datasets, originating from the same university, exhibited more concentrated BP distributions. In contrast, the PPG dataset, derived from public databases, featured a wider BP distribution and increased individual differences in pulse waves due to the

diversity of patient diseases. Furthermore, the model proposed in this study achieved higher accuracy on the iPPG and PPWr datasets compared to existing studies [25] and [27], benefiting from the utilization of three subnets and two strategies.

B. Advantages and Prospects of This Research

This study has two distinct advantages. Firstly, considering that the OVAR-BPnet model achieves high-performance BP estimation in cuff-less BP measurement through feature extraction and BP regression units, as well as data augmentation and regularization strategies utilizing PPG, iPPG, and PPWr signals, this indicates the potential of the model in cuff-less BP measurement. Secondly, to the best of our knowledge, this is the first study to conduct BP estimation experiments on three types of pulse wave datasets. The commendable performance of our model in estimating BP on different types of pulse waves highlights its strong generalizability. This is crucial because different types of pulse waves can provide different insights into cardiovascular health, and the model's ability to handle multiple types is beneficial for real-world applications.

In terms of application prospects, the PWDL method holds great promise. Research into BP estimation via iPPG and PPWr signals remains nascent, marking this study as a significant endeavor [46]. Among them, the experimental outcomes reaffirm the viability of employing pulse diagnostic apparatus for BP self-estimation, providing support for combining traditional Chinese medicine pulse diagnosis with BP measurement to improve diagnostic accuracy [47]. We anticipate applying the PWDL approach to facial and pulse diagnostic apparatus for BP assessments in the future. Furthermore, previous research has shown that utilizing multi-channel or multi-type pulse waves to integrate the unique BP characteristics of various pulse waves of the same individual can produce better results [14]. This insight prompted us to plan to apply the PWDL method to apparatuses capable of collecting multiple types of pulse waves simultaneously to achieve high-precision BP estimation.

C. Limitations of This Research

However, this study also has two shortcomings. First, the proposed BP model is not capable of estimating marginal BP classes. We find that the model's estimation performance for the marginal SBP classes (80 mmHg–105 mmHg and 160 mmHg–180 mmHg) is not as good as for the SBP classes in the major (105 mmHg–160 mmHg). This is attributed to the imbalance in the distribution of BP labels in the dataset, resulting in insufficient model learning for pulse waves with marginal BP classes. However, the distribution of BP labels naturally tends to be unbalanced in actual population cohorts, which poses a challenge to model training [48]. To solve it, future research can innovatively explore advanced methods in the field of ensemble deep imbalance regression to improve the performance of the model in terms of estimation accuracy [49].

The second shortcoming of this study is that the model in the PWDL approach has not yet demonstrated its continuous BP estimation capabilities on a private dataset. The 520 subjects in the public dataset all have labeled data segments over 2

hours. Therefore, the model has the continuous BP estimation capability on PPG data. However, for subjects in the private dataset, only a pulse wave segment and corresponding label are provided for each subject. Looking ahead, we will verify the effectiveness of the OVAR-BPnet model for continuous BP estimation based on iPPG and PPWr data by collecting new data during follow-up of participants in private datasets.

V. CONCLUSION

In this work, we propose a general PWDL approach for BP estimation, introducing the hybrid deep BP model named OVAR-BPnet. The BP model incorporates a single-period waveform feature extraction unit, a multi-period waveform feature extraction unit, and a phenotype-fused BP regression unit, alongside strategies for data augmentation and label sequence regularization. The two feature extraction units fully learn the intrinsic comprehensive features of the pulse wave, while strategies for data augmentation and label sequence regularization further refine the performance of the model. Tested on three types of pulse wave datasets (PPG, iPPG, and PPWr), the proposed BP model exhibited advanced BP estimation performance. The interpretability study visually analyzed the process of relating pulse waves to BP and the pulse wave learning ability of the model. Furthermore, experiments on three types of pulse waves verified the generalizability of the OVAR-BPnet model. In the future, we aim to integrate this model into medical diagnostic systems, marking a significant contribution to intelligent health-care.

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